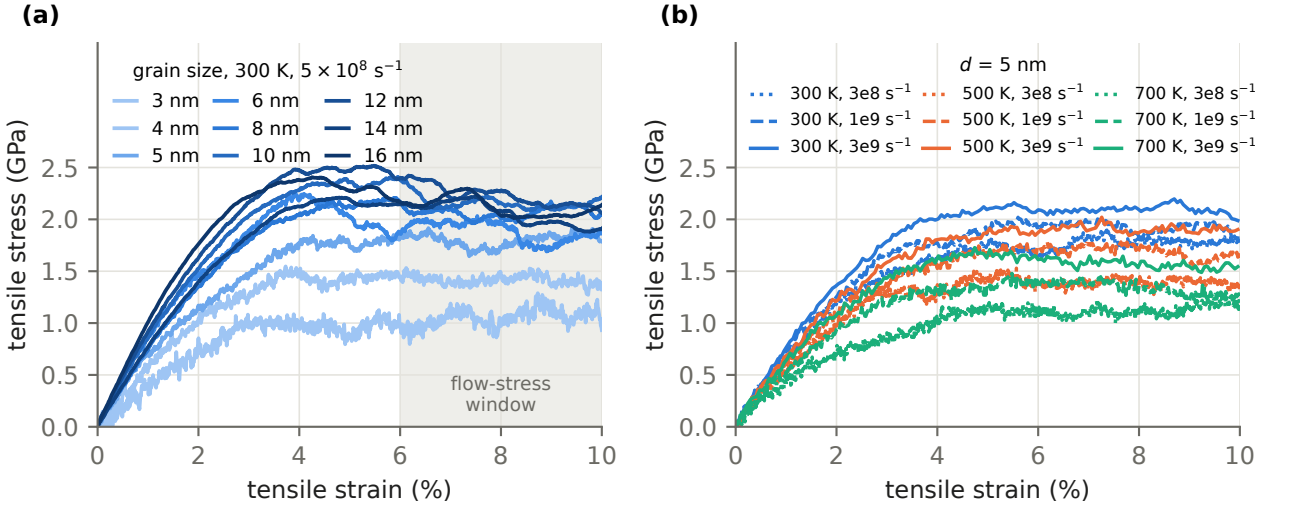

(a)
grain size, 300 K, 5 × 10^8 s^−1
3 nm
4 nm
5 nm
6 nm
8 nm
10 nm
12 nm
14 nm
16 nm
flow-stress window
tensile stress (GPa)
tensile strain (%)
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
10
(b)
d = 5 nm
300 K, 3e8 s^−1
300 K, 1e9 s^−1
300 K, 3e9 s^−1
500 K, 3e8 s^−1
500 K, 1e9 s^−1
500 K, 3e9 s^−1
700 K, 3e8 s^−1
700 K, 1e9 s^−1
700 K, 3e9 s^−1
tensile stress (GPa)
tensile strain (%)